

Assignment 1

From Data to Decision: OLS and the Determinants of Health Costs

How this assignment works. Submit one knitted R Markdown file (text + code). Please include the AI-use statement described in the course AI policy. **There are four blocks.** Blocks A and D are your own thinking and carry the most weight. Block B is where you lean on AI to do the mechanical work. Block C tests whether you can catch what AI gets wrong. Late assignments are penalized 10% per day.

Video (★). One part below is marked with a ★. For that part, submit a video of no more than 3 minutes explaining both how you reached your answer and what it means. See the syllabus for full instructions.

The data. We use the *healthyR_data* administrative dataset (187,721 hospital visits with demographics, charges, payments, and quality metrics), from the *healthyR* R package.

Research question. What is the effect of length of stay (*length_of_stay*) on the total cost of a hospital visit (*total_charge_amount*)? Candidate covariates: place of service (*ip_op_flag*), service line, payer, admission day of week, and expected 30-day readmission.

Block A — Motivation

This is the part AI can draft but cannot own. Write it yourself; you will defend it in Block D.

- a. **Why should anyone care?** In one paragraph, motivate this question from three vantage points — a physician's, a researcher's, and a policy-maker's. What decision would each make differently if they knew the answer?
- b. **Draw a DAG.** Sketch the hypothesized causal relationships among the outcome, length of stay, and the covariates (hand-drawn and scanned is fine). Justify the arrows in text. Then the hard part: *where is this DAG most likely wrong or incomplete, and what would that do to your estimate?*
- c. **State the identifying assumption in one sentence.** Complete this sentence precisely: “A regression of costs on length of stay recovers the causal effect only if ...” Then give one reason to believe it and one reason to doubt it.

Block B — Coding

Here you should **use AI (Copilot, ChatGPT, Claude, etc.) to write the code**. The goal is fluent, well-directed use, not writing every line yourself. For **one** of the parts below, paste into your write-up the prompt you gave the AI, the raw code it returned, and what you did to correct or redirect it. (This is your “prompt + critique” artifact; the AI policy explains it.). *(Do not submit this for a part where there was no redirection required, as that is besides the point of this block.)*

- a. **Clean.** Before running any regressions, you will need to clean your data. Convert all variables to appropriate storage types, and choose how to deal with any missing data in your variables. Construct indicators for (i) inpatient, (ii) outpatient, (iii) payers, (iv) service types, (v) weekend admissions, and (vi) expected readmission within 30 days. For payers and service types, you do

not have to include dummy variables for each individual category, you can (and should!) aggregate these as you see fit. Describe the pipeline your team decided on to clean the data and justify your choices.

- b. **Summarize.** Starting from *healthyR_clean.csv*, produce one well-formatted summary table: means and SDs for continuous variables, counts and percentages for categoricals. Does anything about the sample stand out?
- c. **Visualize.** Make one figure showing the relationship between length of stay and costs that you find compelling. Report it and tell me the story it shows.
- d. **Univariate OLS.** Regress total charges on length of stay alone. Report coefficients, SEs, confidence intervals, p-values, R^2 , and N in a formatted table. How much does one extra week of stay raise costs?
- e. **Multivariate OLS.** Now estimate the full specification implied by your Block A DAG. In doing so, be sure to consider whether any variable needs a transformation, and report robust standard errors of your choice. Add this as a new column to your table. How and why did the length-of-stay coefficient change?

Block C — AI Correction

A student asked an AI assistant to “estimate the effect of length of stay on costs” and pasted the reply below into their submission without checking it. **The code runs without any error message**, and the numbers look plausible. But this analysis has **at least three consequential problems**.

Reproduce the snippet on the real data, then find and fix the problems. For each one: (i) say what is wrong, (ii) show the corrected code or statement, and (iii) explain in one sentence how it changes the conclusion. If you spot an extra problem I did not plant, say so.

The AI-generated analysis:

```
library(tidyverse); library(fixest)
df <- read_csv("healthyR_clean.csv")

# Log-transform the outcome and estimate the model
df <- df %>% mutate(log_cost = log(total_charge_amount))
m <- feols(log_cost ~ length_of_stay + factor(payer) + readmit_30,
            data = df)                # readmit_30 = expected 30-day readmission
summary(m)                          # reports the default standard errors

# Coefficient on length_of_stay is about 0.06, so:
# "Each additional day of stay increases total charges by about $0.06."
```

Block D — Interpret & Defend

- a. ★ **Translate for a decision-maker.** A hospital administrator who cannot read a regression table asks: “So if we cut average length of stay by one day, how much do we save, and how sure are you?” Answer them in plain language *in your video*. Be honest about what your estimate does

and does not establish — in particular, whether it is the causal number they are implicitly asking for, and what would have to be true for it to be.

- b. **Revisit your assumption.** Return to the one-sentence identifying assumption from Block A. After running the regressions, are you more or less worried about it? Provide one piece of evidence (a regression or figure) that speaks to whether length of stay is plausibly “as good as randomly assigned” once you condition on your controls — even if imperfect.